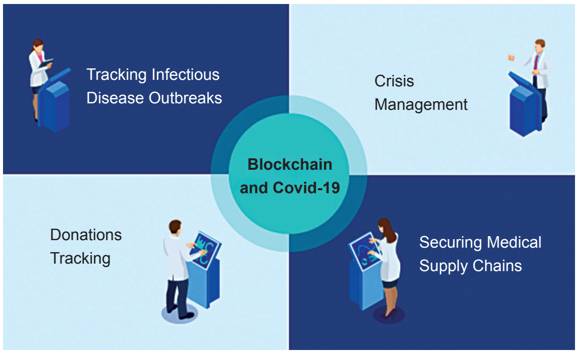
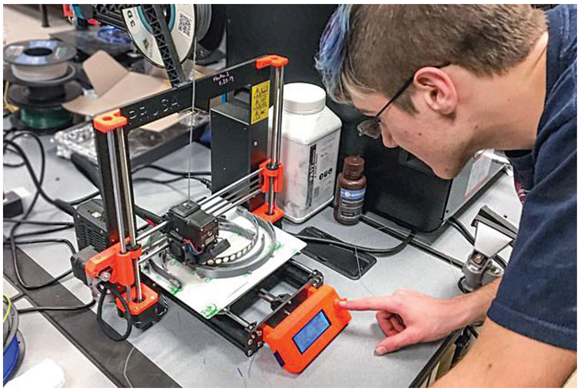
people. This technology has the potential of becoming an integral part of the global response to the coronavirus pandemic by tracking the spread of the disease, maintaining the sustainability of medical supply chains, and managing insurance payments. Its decentralised nature and cryptographic algorithm make it immune to attack and hacking.

***Open source technologies:*** During epidemic outbreaks, rapid data sharing is critical to allow for a greater understanding of the origins and spread of the virus and function as a source for effective prevention, treatment, and care. The capacity of information technologies permits low-cost dissemination and collaboration of data, which enables the creation of a large number of repositories and information technology platforms for data sharing. A substantial number of these activities are coordinated by international organisations such as the World Health Organization (WHO) and European Centre for Disease Prevention and Control. An increasing number of bottom-up, open-data enterprises and open-source projects have been initiated, enabling easy and quick access to research data and scientific publications as well as sharing designs for the production of critical medical equipment such as ventilators and face shields.

***3D printing:*** The existing healthcare system capacity in various countries, planned for the normal times, is limited and unable to meet the rapidly increasing demand for medical hardware such as face masks, ventilators, and breathing filters, etc, required to treat Covid-19 patients.




Blockchain technology has emerged as a key technology in the domain of pandemic management (Credit: *https://www.bbvaopenmind.com*)



A 3D printer being used to print personal protective equipment in response to the lack of safety gear for medical professionals (Credit: *https://www.usda.gov*)

Governments around the world are taking increasingly radical actions to increase production and enhance the supply of the necessary medical equipment. Three-dimensional (3D) printing technology can be of immense use during this emergency in supporting the endeavour of healthcare workers and hospitals in keeping patients alive.

***Nanotechnology:*** The Covid-19 outbreak has tested the limits of healthcare systems, posing serious questions about the adequacy of conventional therapies and diagnostic tools. Quarantine, isolation, and infection-control measures are being used to prevent the spread of the virus and isolate those who show the symptoms.

However, a specific antiviral agent is lacking to treat the infected and decrease viral spreading and transmission. The use of nanotechnology offers new opportunities for the prevention, diagnosis, and treatment of Covid-19. Nanotechnology is broadly defined as the design and application of materials and devices where at least one dimension is less than a hundred nanometres.

The application of nanotechnology in the medical field called 'nanomedicine' includes the use of nanomaterials for diagnosis, treatment, control, and prevention of diseases.

Nanoparticles can be extensively used in the development of better and safer drugs, tissue-targeted treatments, personalised nanomedicines, and early diagnosis as these have unique properties such as small size, improved solubility, surface adaptability, and multi-functionality.

Nanotechnology can help fight against Covid-19 by offering infection-safe personal protective equipment (PPE) to enhance the safety of healthcare workers and develop effective antiviral disinfectants and surface coatings to inactivate the virus and prevent its spread.

Nano-based biosensors could be used in diagnostics for viral infection with high specificity and sensitivity. Nanoparticle-based markers can be used to study the mechanism by which viruses infect host cells. A new generation of vaccines could be developed based on nanomaterials with improved antigen stability, target delivery, and controlled release.

Contaminated surfaces in public places, such as hospitals, parks, public transportation, and schools, are a common source for outbreaks of infection. Nano-based surface coatings have the potential for the prevention of such infections.

***Drones and robots:*** Drones are being deployed for disinfection and street patrols for food and medicine